problem 2 (Brain and Graph Algorithms)

(a) Hierarchial feed forward cascades

Algorithm:

Topological Sort (or DAG shortest paths)

Reason:

Information flows forward with no loops → this is a Directed Acyclic Graph (DAG)

Time complexity: $O(V+E)$

(b) Recurrent feedback circuits

Algorithm:

Strongly Connected Components (SCC) e.g Kosaraju or Tarjan

Reason: Feedback loops mean nodes form cycles → SCC finds these groups.

Time complexity: $O(V+E)$

(c) Fastes reactive pathways

Algorithm: Dijkstra's Algorithm

Reason: we want the shortest (minimum delay) path in a weighted graph.

Time complexity: $O(V+E) \log V)$

Limitation of static graph algorithms

A key limitation is that the brain is dynamic, but graphs are static. Edge weight and connections can change over time.

Real thinking involves time-dependent processes

So static algorithms cannot fully capture real neural activity.

problem 1 (Cut Property Proof)

claim:

For any cut $(S, V \setminus S)$, the minimum-weight edge crossing the cut is in every Minimum Spanning Tree (MST)

Proof (by induction on number of edges):

Base case:

If the graph has only on edge, then it is the minimum edge and it must be in MST. so the statement is true.

Inductive step: Assume the statement is true for graphs with fewer edges.

Now consider a graph G, Let e be the minimum-weight edge crossing the cut.

suppose for contradiction that e is not is some MST T.

If we add e to T, a cycle is formed.

In this cycle, there must be another edge e' that also crosses the cut.

Since e is the minimum edge, we have: $w(e) \leq w(e')$

If we remove e' and keep e, we get a new spanning tree with equal or ~~sa~~ smaller weight.

So T was not minimal $\rightarrow$ contradiction

Therefore, e must belong to every MST.